

EESC4550/5550: Practical Atmospheric Modeling (Fall 2026)

Lab 1 – Linux Computing and Command Line Operations

Total Score: 15

Learning Objectives:

- Reinforce basic Linux commands and concepts (navigation, file operations, content viewing, searching, file permission management, and process management).
- Learn to use symbolic links and environment variables.
- Inspect and extract information from meteorological NetCDF files using `ncdump`.
- Write and submit a simple Slurm batch job.

Pre-requisites:

- You are familiar with the concepts in the lecture.
- You will need your computing ID (e.g. s1155xxxxxx, 1155xxxxxx).

Part 0. Setup

Connecting to the central cluster:

```
ssh <your-computing-id>@chpc-login.itsc.cuhk.edu.hk
```

Go to your course project folder, and copy the starter directory for this lab:

```
cd /lustre/project/eesc4550/<your-computing-id>
cp -r /lustre/project/eesc4550/lab1 .
```

All the files graded for this lab should then be located under the path

`/lustre/project/eesc4550/<your-computing-id>/lab1/`.

Important: Please work on and save your scripts under the `/scripts` directory.

Part 1. Basic File and Directory Operations

Question 1 (1 pt). Create the following directory structure inside `lab1/`.

```
lab1/
├── data/
│   ├── raw/
│   └── processed/
├── files/
├── logs/
└── scripts /
```

Inside `data/raw/`, create the following files using brace expansion:

- `obs_001.dat` through `obs_031.dat`
- `file_a.txt`, `file_b.txt`, `file_c.txt`

Make sure that you have exactly 31 `.dat` files and 3 `.txt` files.

List **all** files and directories (including hidden) inside `lab1/`.

Question 2 (0.5 pts). Let `D` be the **last digit of your Computing ID** (e.g., if you are `s1155123456`, then `D = 6`). Use a single wildcard pattern to list **only** the `.dat` files in `data/raw/` whose number ends in `D`. Save the output to `data/processed/wildcard_list.txt`. Show the contents of this file.

Question 3 (0.5 pts). Inside `data/processed/`, create one empty file: `summary.txt`. Write the following header line into `data/processed/summary.txt` using a single command:

```
Timestamp, Ozone (DU), Temp (K)
```

Then append the line `"298, 1013, 30"` to the same file and a third line containing your Computing ID (e.g., `s1155xxxxxx` or `1155xxxxxx`). Show the file contents of `summary.txt`.

Question 4 (1 pt). Using a single `sed` command with a regular expression, replace the header to:

```
Timestamp,Ozone (DU),Temp (K)
```

Show the file contents so that we know you have operated correctly.

Question 5 (0.5 pts). List **all** files and directories (including hidden) inside `data/` and its subdirectories and save the output to `logs/dir_list_old.log`. Then, copy all `.txt` files from `data/raw/` to `data/processed/`.

Question 6 (0.5 pts). Remove the three `.txt` files in `data/raw/`. List **all** files and directories (including hidden) inside `data/` and its subdirectories. After the removal, save the output to `logs/dir_list_new.log`.

Part 2. Viewing and Searching File Content

In this part, you will focus on

```
/lustre/project/eesc4550/lab1_supp_files/ozone_station_data.csv.
```

Question 7 (1 pt). Display only the lines that contain `"26"` and save the output to `logs/ozone_26.log`. Then, count how many lines contain the word `"HKG"` (case-insensitive). Extract **only the Hong Kong** (HKG) data. Then, write the count and the extracted data to `logs/hkg_ozone_data.log`. Show the contents of the `.log` files.

Question 8 (1 pt). Use `awk` to compute the **average ozone concentration**, save the number to `data/processed/avg_ozone.txt`. Show the first 2 lines and the last 2 lines of `ozone_station_data.csv`, and combine them into a single file called `data/processed/edges.txt`. Show the file contents.

Part 3. Conditional Statements and Loops

Question 9 (1.5 pts). Complete the empty function bodies in the script `prime_check.sh` that decides whether a number is prime.

Typo found on 2026-09-21: In the function description of `prime_check.sh`, return `0` for numbers `<2`.

Part 4. Environment Variables

Question 10 (1 pt). Complete the script `env_demo.sh`, but instead of hardcoding "Bob" and "81", it derives `NAME` and `AGE` from your Computing ID.

Derivation rule:

- `NAME = $(whoami)` — your full Computing ID.
- `AGE = (sum of all digits in your Computing ID) % 60 + 1` — an integer between 1 and 60.

Part 5. File Permission Management

Question 11 (1 pt).

(a) Create a script `scripts/hello.sh` containing exactly:

```
#!/bin/bash
echo "Hello from $USER"
```

Make it executable **for the owner only** using symbolic mode (`chmod u+x`). Verify with `ls -l /scripts/hello.sh`.

(b) Set the permissions of `data/processed/` to 750 using numeric mode. This means: owner can read/write/execute, group can read/execute, others have no access. Verify with `ls -ld data/processed/`.

(c) Remove **write** permission for the group on `data/processed/summary.txt` using symbolic mode. Verify with `ls -l data/processed/summary.txt`.

Save the three `ls` outputs (with a one-line comment before each) to `logs/file_permissions.log`. Show its contents.

Part 6. Process Management

Question 12 (1 pt). Run the binary file `script/trap_test.sh` in the background. The program writes its PID to `trap_test.pid`. Kill the suspended process **gracefully**. Then, verify that the trap wrote the flag file `flag_<your_id>.txt`. Display its contents with `cat flag_*.txt`. In `scripts/lab1.sh`, fill in the corresponding parts with the PID you found, and the contents of the flag file.

Part 7. Symbolic Link Creation and NetCDF File Inspection with `ncdump`

Question 13 (0.5 pts). A full year of GEOS-Chem diagnostics can easily exceed hundreds of gigabytes. Create one symbolic link per file in the shared folder `/lustre/project/eesc4550/lab1_supp_files`, all inside `data/raw/`. Verify that the links are created correctly by listing them (`.nc4` files only) using the command `ls` with the `-l` option and save the output to `data/processed/nc4_list.txt`.

Question 14 (2.5 pts). Write a Bash script called `extract_metadata.sh` that does the following:

- Loops over **every** `.nc4` file listed in `data/processed/nc4_list.txt`.
- For each file, run `ncdump -h` and **pipe** the output to `grep` to find:
 - The **number of time steps** (look for `time = UNLIMITED ; // (X currently)`).

- The **names of the last 3 variables** (look for `float` or `double` declarations).
- Appends a CSV line to `data/processed/metadata_summary.csv` in the format:
`filename, time_steps, var1, var2, var3`.
- Show the contents of the csv file in a table format.

Part 8. Writing and Submitting a Slurm Batch Job

Question 15 (1.5 pts). We estimate π with Monte Carlo on 10 CPUs. Write `mc.slurm` that:

- Names the job `mc_pi`
- Writes stdout to `logs/mc_%j.out`
- Requests 1 task, 10 CPUs per task
- Executes `python3 mc_pi.py`

Then, provide the output of `mc_pi.py` in `scripts/lab1.sh`, which contains the number of CPUs used, wall time, and your estimated π value.

Checking Your Results:

After running the scripts, your directory should look like the one given below.

```
lab1/
├── data/
│   ├── processed/
│   │   ├── avg_ozone.txt
│   │   ├── env_output.txt
│   │   ├── edges.txt
│   │   ├── log_a.txt
│   │   ├── log_b.txt
│   │   ├── log_c.txt
│   │   ├── metadata_summary.csv
│   │   ├── nc4_list.txt
│   │   ├── prime_output.txt
│   │   ├── summary.txt
│   │   └── wildcard_list.txt
│   └── raw/
│       ├── obs_001.dat
│       ├── obs_002.dat
│       ├── ...
│       ├── obs_031.dat
│       ├── GEOSChem.AdvFluxVert.20190701_0000z.nc4
│       ├── GEOSChem.AerosolMass.20190701_0000z.nc4
│       ├── GEOSChem.Aerosols.20190701_0000z.nc4
│       ├── GEOSChem.DryDep.20190701_0000z.nc4
│       ├── GEOSChem.JValues.20190701_0000z.nc4
│       ├── GEOSChem.SpeciesConc.20190701_0000z.nc4
│       ├── GEOSChem.StateMet.20190701_0000z.nc4
│       ├── geoschem_species_metadata.yml
│       └── ozone_station_data.csv
├── logs/
│   ├── dir_list_new.log
│   ├── dir_list_old.log
│   ├── file_permissions.log
│   ├── hkg_ozone_data.log
│   ├── mc_<jobid>.out          # this might be deleted by the script lab1.sh, but it is okay
│   ├── mc_<jobid>.err          # same as above, just remember to record the output!!
│   └── ozone_26.log
├── scripts/
│   ├── env_demo.sh
│   ├── extract_metadata.sh
│   ├── flag_<your_id>.txt
│   ├── gcc.env
│   ├── hello.sh
│   ├── lab1.sh
│   ├── mc_pi.py
│   ├── mc.slurm
│   ├── prime_check.sh
│   ├── trap_test
│   └── trap_test.pid
└── submission.sh
```

After running `lab1.sh`, we expect to see the output as below (with some information omitted).

```
[b169195@chpc-logina scripts]$ bash lab1.sh
Student ID: b169195      (last digit = 5)

Currently Loaded Modulefiles:
  1) intel/2019u4  2) cmake/3.18.2
#### Question 1 (1 pt). Files and Directory Structure
..
.  .. data logs scripts submission.sh
<lines omitted>
#### Question 2 (0.5 pt). Wildcard Listing
<lines omitted>
#### Question 3 (0.5 pt). summary.txt
Timestamp, Ozone (DU), Temp (K) Token
298, 1013, 30
s1155xxxxxx
#### Question 4 (1 pt). Clean the Header with sed
Timestamp,Ozone (DU),Temp (K) Token
298, 1013, 30
s1155xxxxxx
#### Question 5 (0.5 pt). Copy .txt Files
data/:
.
..
processed
raw

data/processed:
.
..
summary.txt
wildcard_list.txt
<lines omitted>
#### Question 6 (0.5 pt). Remove .txt Files
data/:
.
..
processed
raw

data/processed:
.
..
log_a.txt
log_b.txt
log_c.txt
summary.txt
wildcard_list.txt
<lines omitted>
#### Question 7 (1 pt). Search, Count, Extract
2025-01-06,HKG,26.1
<lines omitted>
<a number with the count>
2025-01-01,  HKG ,25.5
<lines omitted>
#### Question 8 (1 pt). Average + Edges
Average O3: <a number with the averaged ozone conc>
Date, Station , O3_ppb
2025-01-01,  HKG ,25.5
<lines omitted>
#### Question 9 (1.5 pts). prime_check.sh
<lines omitted>
#### Question 10 (1 pt). env_demo.sh
<lines omitted>
#### Question 11 (1 pt). File Permissions
<lines omitted>
#### Question 12 (1 pt). Trap Program
PID=<value>
START=<value>
KILL=<value>
HASH=<string>
#### Question 13 (0.5 pt). nc4_list.txt
lrwxrwxrwx. 1 b169195 b169195 80 Sep 18 23:22 data/raw/GEOSChem.AdvFluxVert.20190701_0000z.nc4 ->
/lustre/project/eesc4550/lab1_supp_files/GEOSChem.AdvFluxVert.20190701_0000z.nc4
lrwxrwxrwx. 1 b169195 eesc4550 80 Sep 18 23:22 data/raw/GEOSChem.AerosolMass.20190701_0000z.nc4 ->
/lustre/project/eesc4550/lab1_supp_files/GEOSChem.AerosolMass.20190701_0000z.nc4
<lines omitted>
data/raw/GEOSChem.AdvFluxVert.20190701_0000z.nc4
data/raw/GEOSChem.AerosolMass.20190701_0000z.nc4
<lines omitted>
#### Question 14 (2.5 pts). extract_metadata.sh


| filename                                | time_steps | var1       | var2      | var3           |
|-----------------------------------------|------------|------------|-----------|----------------|
| GEOSChem.AdvFluxVert.20190701_0000z.nc4 | 2          | P0         | AREA      | AdvFluxVert_O3 |
| GEOSChem.AerosolMass.20190701_0000z.nc4 | 2          | AerMassHMS | AerMassBC | AerMassASOA    |


<lines omitted>
#### Question 15 (1.5 pts). Submit mc.slurm
CPUs: 10
Wall: <value>s
PI estimate: <value>

=== End of Lab 1. ===
```

Submission Guidelines and Deliverables:

To submit your solution for this lab, run `bash scripts/submission.sh` from inside your `lab1/` folder. This produces two files in the current directory, namely `lab1_<your_computing_id>.tar.gz` and `lab1_<your_computing_id>.tar.gz.md5`. Upload **both** files to the course submission system. The autograder prepared by the TA will report your score for each task. Please ensure that all files are in the correct locations. Otherwise, the autograder may report a score of **0**. We are aware that students may try to fool the autograder, the teaching team is aware of this and has made substantial effort to prevent this issue. Manual checks are performed if necessary.

Viewing .nc4 Files:

To view only the metadata and structural header of a netCDF4 file on Linux, we can use `ncdump -h`. For example, `ncdump -h GEOSChem.AdvFluxVert.20190701_0000z.nc4 | less` extracts the dimensions, variables, and attributes without printing any of the actual data values, and it is then piped into `less` so you can scroll through it comfortably.

To view the data of a specific variable, we can use the command `ncdump -v your_variable_name your_file.nc4 | grep -A 20 "data:"`. Here, the data section always starts after the word `data:`, and hence we use `grep` with `-A` after to jump straight to the variable's values.

Nevertheless, the netCDF utility path is not loaded in the default environment. To solve this issue, we can "source" a specific environment file `gcc.env`, which loads the netCDF utility to our `PATH`.

----- End of Lab 1 -----

Note: If you have utilized any AI tools in this lab assessment, you are obliged to attach an explicit citation or a proper acknowledgement of the use of these AI tools.

On AI Use in Preparing This Manual:

Portions of this lab manual, including the structure of `lab1.sh`, and some of the helper scripts, were drafted with the assistance of an AI language model (**DeepSeek-V3**), and subsequently reviewed, tested, and edited by the teaching team. The scientific content, command sequences, and grading criteria were verified by running the scripts on the CUHK cluster before release.